INDEX

INSTITUTE

VISION MISSION

INSTITUTE VISION AND MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.

DEPARTMENT

VISION
MISSION

DEPARTMENT

VISION AND MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character

PROGRAM EDUCATIONAL OBJECTIVES, PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES

Program Educational Objectives (PEOs)

Graduates of this programme will :

PEO 1: Adapt to evolving technology.

PEO 2: Provide optimal solutions to real time problems.

PEO 3: Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSO s):

A graduate of the Computer Science and Engineering Program will be able to:

PSO 1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

PSO 2: Use Professional engineering practices and strategies for development and maintenance of software. [K3]

Program Outcomes (POs):

Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge**: Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. **[K3]**
2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. **[K4]**
3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. **[K5]**
4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. **[K5]**
5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations**.** **[K3]**
6. **The Engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. **[K3]**
7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. **[K3]**
8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. **[K3]**
9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. **[K6]**
10. **Communication**: Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. **[K2]**
11. **Project management and finance**: Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. **[K6]**
12. **Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. **[K1]**

ACADEMIC CALENDAR

Grams: "TECHNOLOGY"
Email: dapjntuk@gmail.com

Phone: 0884-2300991
Mobile: +9963993504

Directorate of Academic & Planning
JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA
KAKINADA-533003, Andhra Pradesh, INDIA
(Established by AP Government Act No. 30 of 2008)

Lr. No. JNTUK/DAP/AC/B. Tech/III Year/2019-20 *Date: 30-05-2019*

Dr. A. Mallikarjuna Prasad
M.E, Ph.D.,
Director, Academic Planning

To
All the Principals of Affiliated Colleges,
JNTUK, Kakinada

ACADEMIC CALENDAR FOR B.TECH III YEAR (2017 BATCH)

Description	From	To	Weeks
I SEMESTER			
Commencement of Class Work	**10.06.2019**		
I Unit of Instructions	10.06.2019	03.08.2019	8W
I Mid Examinations	05.08.2019	10.08.2019	1W
II Unit of Instructions	12.08.2019	05.10.2019	8W
II Mid Examinations	07.10.2019	12.10.2019	1W
Preparation & Practicals	14.10.2019	19.10.2019	1W
End Examinations	21.10.2019	02.11.2019	2W
Commencement of II Semester Class Work	**18.11.2019**		
II SEMESTER			
I Unit of Instructions	**18.11.2019**	11.01.2020	8W
I Mid Examinations	13.01.2020	23.01.2020	1W
II Unit of Instructions	24.01.2020	21.03.2020	8W
II Mid Examinations	23.03.2020	28-03-2020	1W
Preparation	30.03.2020	04.04.2020	1W
End Examinations	06.04.2020	18.04.2020	2W
Commence of IV Year Class Work	**08.06.2020**		

A.M.prasad
Director Academic Planning

Copy to the Secretary to the Hon'ble Vice Chancellor, JNTUK.
Copy to PA to the Rector, JNTUK.
Copy to PA to the Registrar, JNTUK.
Copy to PA to the Director of Evaluation, JNTUK.

CO-CURRICULAR AND EXTRA CURRICULAR ACTIVITIES

05/06/2019 – World Environment Day	**15/09/2019 – Engineers Day**		**21/03/2019 – International Forest Day**
21/08/2019 to 24/08/2019– 1st year B.Tech. Introduction program	**16/09/2019 – World Ozone Day**	**22/12/2019 – National Mathematics Day**	**22/03/2019 –World Water Day**
15/08/2018 – Independence Day	**September – Intramurals**	**26/01/2019 – Republic Day**	**23/03/2019 – World Meteorological Day**
05/09/2019 – Teachers Day	**11/11/2019 – National Education Day**	**08/03/2019 – International Women’s Day**	**24/03/2019 – Earth Hour**
	03/12/2019 – Antipollution Day		**In the month of March – Association Days**

SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science &Engineering (Accredited by NBA)

CLASS CONSOLIDATED TIME TABLE

Class: III B.Tech II Semester w.e.f. 18-11-2019

Sec :A Class Coordinator :**Mrs.N. Hiranmayee** Room No : B-401

Periods	1	2	3	4		5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)	1:00PM 2:00PM	(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	STM	**DWM LAB**			**Lunch Break**	CN	**Technical Training**	
Tue	DAA	**APPTITUDE**		CS		DWM	**Technical Training**	
Wed	CN	**ECS**	DAA	CS		STM	**Technical Training**	
Thu	CS	DAA	DWM	IPR		**NP LAB**		
Fri	DWM	STM	CS	DAA		STM	CN	IPR
Sat	DWM	CN	**Technical Training**			**ST LAB**		

Class: III B.Tech II Semester w.e.f. 18-11-2019

Sec :B Class Coordinator :**Mr.M S Kumar Reddy** Room No : B-402

Periods	1	2	3	4		5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)	1:00PM 2:00PM	(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	CN	CS	**Technical Training**		**Lunch Break**	**APPTITUDE**		DMW
Tue	STM	DWM	CN	DAA		**ST LAB**		
Wed	DWM	DAA	STM	CN		**NP LAB**		
Thu	IPR	**ECS**	DAA	STM		CS	**Technical Training**	
Fri	DAA	**DMW LAB**				CS	**Technical Training**	
Sat	CS	STM	IPR	DMW		CN	**Technical Training**	

Class: III B.Tech II Semester w.e.f. 18-11-2019

Sec :C Class Coordinator : **Mr. N.V.Murali Krishna Raja** Room No : B-301

Periods	1	2	3	4		5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)	1:00PM 2:00PM	(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	CS	DWM	**Technical Training**		**Lunch Break**	**ST LAB**		
Tue	STM	CS	DWM	DAA		CN	**APPTITUDE**	
Wed	DAA	**DWM LAB**				STM	**ECS**	CS
Thu	STM	CN	IPR	DAA		DWM	**Technical Training**	
Fri	IPR	DAA	STM	DWM		CN	**Technical Training**	
Sat	CN	**NP LAB**				CS	**Technical Training**	

Class :III B.Tech II Semester w.e.f.18-11-2019

Sec :D Class Coordinator : **Mrs. D. AnjaniSuputri Devi** Room No : B-302

Periods	1	2	3	4		5	6	7
Time / Day	(09.30 AM-10.30 AM)	(10.30 AM-11.20 AM)	(11.20 AM-12.10 PM)	(12.10 PM-01.00 PM)	1:00PM 2:00PM	(02.00 PM-02.50 PM)	(02.50 PM-03.40 PM)	(03.40 PM-04.30 PM)
Mon	DMW	**ECS**	CN	STM		CS	**Technical Training**	
Tue	DAA	**NP LAB**				DMW	**Technical Training**	
Wed	CS	**APPTITUDE**		DAA		CN	**Technical Training**	
Thu	DAA	**DMW LAB**				CS	DMW	STM
Fri	CN	IPR	DAA	CS		**ST LAB**		
Sat	STM	DMW	**Technical Training**			STM	CN	IPR

Staff Details:

S. No.	Course Code	Course Name	Section			
			A	B	C	D
1.	R1632051	CN	Mrs. D. AnjaniSuputri	Mr. M S Kumar Reddy	Mr. M S Kumar Reddy	Mrs. D. AnjaniSuputri
2.	R1632052	DWM	Dr. V.S.Naresh	Mrs. G. Loshma	Dr. D. Jaya Kumari	Mrs. G. Loshma
3.	R1632053	DAA	Dr. O. Sri Nagesh	Dr.V.Venkateswara Rao	Dr. O. Sri Nagesh	Dr.V.Venkateswara Rao
4.	R1632054	STM	Mrs. N. Hiranmayee	Mrs. N. Hiranmayee	Mr. N.V.Murali Krishna Raja	Mr. P Sireesha
5.	R1632055C	CS	Mr. S Sathappan	Mr. G.Sriram Ganesh	Mr. B.Kiran Kumar	Mr.B.Kiran Kumar
6.	R1632056	NP LAB	Mrs. D. AnjaniSuputriDevi, Dr. S. P. Malarvizhi, Mrs. A. Lakshmi Lavanya	Mr. M S Kumar Reddy, Dr. S. P. Malarvizhi, Mrs. A. Lakshmi Lavanya	Mr. M S Kumar Reddy, Mrs. D. SasiRekha, Dr. J. Veeraraghavan	Mrs. D. AnjaniSuputri Devi, Mrs. A. Lakshmi Lavanya, Mrs I Shanthi
7.	R1632057	ST LAB	Mrs. N. Hiranmayee Mrs. A.Leelavathi Mr. S Sathappan	Mrs. N. Hiranmayee Mrs. A.Leelavathi Mr. S Sathappan	Mr. N.V.Murali Krishna Raja Mrs. A.Leelavathi	Mrs. P Sireesha Mrs. D. SasiRekha Mr. S Sathappan
8.	R1632058	DWM LAB	Dr. V.S.Naresh Mrs. D. SasiRekha Mr. M Vamsi Krishna	Mrs. G. Loshma Mrs. D. AnjaniSuputri Devi Mrs. M Anantha	Dr. D. Jaya Kumari Mr. M Vamsi Krishna Mr. B.Kiran	Mrs. G. Loshma Mrs. P Sireesha Mr. B.Kiran Kumar
9.	R1632049	IPR	Mr. N.V.M K Raja	Mrs. A. Lakshmi Lavanya	Mrs. A. Lakshmi Lavanya	Mr. A.Rajesh
10.		TT	Dr.V.Venkateswara Rao / Dr.O Sri Nagesh	Dr.V.Venkateswara Rao / Dr.O Sri Nagesh	Dr.V.Venkateswara Rao / Dr.O Sri Nagesh	Dr.V.Venkateswara Rao / Dr.O Sri Nagesh
11.		ECS	Dr.T.SUJANI	Mr.SURESH.T	Ms.K.V.L.B. DEVI	Ms.K.V.L.B. DEVI
12.		Aptitude	J N V SOMAYAJULU	J N V SOMAYAJULU	J N V SOMAYAJULU	J N V SOMAYAJULU

Head of the Department

COURSE STRUCTURE

III Year - II Semester

S. No.	Subjects	L	T	P	Credits
1	Computer Networks	4	2	- -	3
2	Data Warehousing and Mining	4	- -	- -	3
3	Design and Analysis of Algorithms	4	- -	- -	3
4	Software Testing Methodologies	4	- -	- -	3
5	**Open Elective:** i. Artificial Intelligence i. Internet of Things iii Cyber Security iv.Digital Signal Processing v.Embbeded Systems vi. Robotics	4	- -	- -	3
6	Network Programming Lab	--	- -	3	2
7	Software Testing Lab	--	- -	3	2
8	Data Warehousing and Mining Lab	--	- -	3	2
9	IPR & Patents	--	2	- -	--
10	English Communication Sills				
11	Aptitude				
	Total Credits				**21**

LESSON PLANS

Computer Networks

LESSON PLAN

Academic Year: 2019-20	**Programme: B.Tech**
Year/ Semester: III/II	**Section: A,B , C & D**
Name of the Course: Computer Networks	**Course Code:C-310**

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO.1 Describe Network Protocols and Architectures. [K2]
CO.2 Discuss Fundamental Concepts Of Physical Layer Functionalities. [K2]
CO.3 Demonstrate Data Link Layer Functionalities. [K3]
CO.4 Develop MAC Layer Protocols. [K3]
CO.5 Illustrate Concepts of Network Layer And Routing Algorithms. [K3]
CO.6 Demonstrate Transport Layer And Application Layer Protocols. [K3]

TEXT BOOKS:

1. Tanenbaum and David J Wetherall, Computer Networks, 5th Edition, Pearson Edu, 2010
2. Computer Networks: A Top Down Approach, Behrouz A. Forouzan, FirouzMosharraf, McGraw Hill Education.

REFERENCE BOOKS:

1. Larry L. Peterson and Bruce S. Davie, "Computer Networks - A Systems Approach" (5th ed), Morgan Kaufmann/ Elsevier, 2011.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	70	70
2	65	60
3	65	60
4	60	60
5	60	60
6	60	60

CO 1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of Vision, Mission of the Dept.and PEOs,Pos,& PSOs of the Programme		1	Lecture	BB
2		Describe Network Topologies WAN, LAN, MAN.	K1	1	Lecture	BB
3		Describe Reference models- The OSI Reference Model	K1	2	Lecture	BB
4		Discuss TCP/IP Reference Model	K2	1	Lecture With Discussion	BB
5		Differentiate OSI and TCP/IP Reference Models	K2	1	Lecture	BB

CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Describe Physical Layer – Fourier Analysis	K1	1	Lecture	BB
2		Explain The Maximum Data Rate of a Channel - Guided Transmission Media	K2	2	Lecture with Discussion	BB
3		Describe Bandwidth Limited Signals – Digital Modulation and Multiplexing: Frequency Division Multiplexing	K1	2	Lecture with Discussion	BB
4		Discuss Time Division Multiplexing, Code Division Multiplexing	K2	1	Lecture	BB+ICT
5		Explain Data Link Layer Design Issues, Error Detection and Correction	K2	2	Lecture with Discussion	BB+ICT
6		Discuss Elementary Data Link Protocols, Sliding Window Protocols	K2	2	Lecture with Discussion and in class Assignment	BB+CRE

CO 3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Describe The Data Link Layer - Services Provided to the Network Layer	K1	1	Lecture	BB
2		Illustrate Framing – Error Control – Flow Control, Error Detection and Correction – Error-Correcting Codes – Error Detecting code	K3	3	Lecture	BB
3		Explain Elementary Data Link Protocols- A Utopian Simplex Protocol	K2	1	Lecture with Discussion	BB
4		Discuss A Simplex Stop and Wait Protocol for an Error free channel-A Simplex Stop and Wait Protocol for a Noisy Channel.	K2	2	Lecture	BB+ICT
5		Demonstrate Sliding Window Protocols-A One Bit Sliding Window Protocol	K3	2	Lecture with Discussion	BB+ICT
6		Demonstrate A Protocol Using Go-Back-NA Protocol Using Selective Repeat.	K3	2	Lecture with Discussion	BB+ICT

CO 4

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Describe The Medium Access Control Sublayer-The Channel Allocation Problem-Static Channel Allocation.	K1	1	Lecture	BB
2		Demonstrate Assumptions for Dynamic Channel Allocation.	K3	2	Lecture with Discussion	BB
3		Discuss Multiple Access Protocols-AlohaCarrier Sense Multiple Access Protocols-Collision-Free Protocols-Limited Contention Protocols.	K2	3	Lecture with Discussion	BB

4		Demonstrate Wireless LAN Protocols, Ethernet-Classic Ethernet Physical Layer-Classic Ethernet MAC Sublayer Protocol.	K3	2	Lecture with Discussion	BB+ICT
5		Demonstrate Ethernet Performance-Fast Ethernet Gigabit Ethernet-10-Gigabit EthernetRetrospective on Ethernet	K3	3	Lecture with Discussion	BB+ICT
6		Explain Wireless Lans-The 802.11 Architecture and Protocol Stack-The	K2	1	Lecture with Discussion	BB+ICT
7		Discuss 802.11 Physical Layer-The802.11 MAC Sublayer Protoco	K2	1	Lecture with Discussion	BB+ICT
8		Discuss The 805.11 Frame Structure-Services.	K2	1	Lecture with Discussion	BB+ICT

CO 5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Describe Design Issues-The Network Layer Design Issues – Store and Forward Packet Switching.	K1	2	Lecture	BB
2		Discuss Services Provided to the Transport layer- Implementation of Connectionless Service-Implementation of Connection Oriented Service.	K2	2	Lecture with Discussion	BB
3		Explain Comparison of Virtual Circuit and Datagram Networks, Routing Algorithms-The Optimality principle.	K2	2	Lecture	BB
4		Compute Shortest path Algorithm, Congestion Control AlgorithmsApproaches to Congestion Control-Traffic	K3	3	Lecture with Discussion	BB

		Aware Routing-Admission Control-Traffic Throttling.				
5		Discuss Load Shedding	K2	1	Lecture with Discussion	BB

CO 6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Describe Transport Layer – The Internet Transport Protocols: Udp, the Internet Transport Protocols: Tcp.	K1	1	Lecture	BB
2		Discuss Application Layer – The Domain Name System: The DNS Name Space, Resource Records.	K2	2	Lecture with Discussion	BB
3		Explain Name Servers.	K2	2	Lecture with Discussion	BB
4		Show Electronic Mail: Architecture and Services.	K3	1	Lecture with Discussion	BB
5		Show The User Agent, Message Formats.	K3	2	Lecture with Discussion	BB
6		Explain Message Transfer, Final Delivery.	K2	1	Lecture with Discussion	BB

Total No. of Classes: 60

Data Warehousing and Mining

LESSON PLAN

Batch: 2017-21	
Academic Year : 2019-20	**Programme:B.Tech**
Year/ Semester: III /II	**Section:A,B,C,D**
Name of the Course: Data Warehousing and Mining	**Course Code: R1632052/ 311**

Course Outcomes (Along with knowledge level):
At the end of the course, the student will be able to

311.1	Explain the concept of Data Mining and its functionalities.[K2]
311.2	Describe Data Preprocessing Techniques like data cleaning, integration, transformation and reduction.[K2]
311.3	Illustrate various Classification Techniques. [K3]
311.4	Demonstrate Alternative techniques for Classification.[K3]
311.5	Demonstrate Association analysis techniques for generating association rules from data.[K3]
311.6	Use different Clustering techniques to cluster data.[K3]

TEXT BOOKS:
1. Data Mining concepts and Techniques, 3/e, Jiawei Han, Michel Kamber, Elsevier.
2. Introduction to Data Mining: Pang-Ning Tan & Michael Steinbach, Vipin Kumar, Pearson.

REFERENCE BOOKS :
1. Data Mining Techniques and Applications: An Introduction, Hongbo Du, Cengage Learning.
2. Data Mining : Introductory and Advanced topics : Dunham, Pearson.

Targeted Proficiency Level and Attainment level (For each course Outcome):

Course Outcome	Targeted Proficiency Level	Targeted Attainment Level
CO1	60	65
CO2	60	60
CO3	60	60
CO4	60	60
CO5	60	60
CO6	60	60

UNIT - I

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO1	**Dissemination of Department Vision, Mission, PEOs, POs, PSOs, COs, Introduction**: Identify what motivated Data Mining.	K1	1	Lecture	BB
2.		State the importance of Data Mining. Describe kinds of Data on which Data Mining can be done.	K1	2	Lecture	BB
3.		Illustrate Data Mining Functionalities.	K2	2	Lecture+ discussion	BB
4.		Illustrate Major Issues in Data Mining	K1	1	Lecture	BB
5.		Explain Attribute Types and Basic Statistical Descriptions of Data	K2	2	Lecture + discussion	BB
6.		Illustrate Data Visualization techniques.	K2	1	Lecture + discussion	BB
7.		Describe Data Similarity and Dissimilarity measures	K2	2	Lecture + discussion	BB
		Total		**11**		

UNIT - II

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO2	**Data Pre-processing**: Identify reasons for pre-processing the data.	K1	1	Lecture	BB
2.		Describe Descriptive Data Summarization.	K2	1	Lecture + discussion	BB
3.		Explain Data Cleaning techniques.	K2	2	Lecture + discussion	BB
4.		Discuss Data Integration and Data Transformation techniques	K2	2	Lecture + discussion	BB
5.		Explain Data Reduction	K2	2	Lecture	BB

		techniques.			+ discussion	
6.		Discuss Data Discretization and Concept Hierarchy Generation.	K2	2	Lecture + discussion	BB
		Total		**10**		

UNIT - III

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO3	**Classification** : Describe the basic concepts of Classification	K1	1	Lecture	BB
2.		Illustrate the approach for solving a classification problem.	K2	1	Lecture + discussion	BB
3.		Explain the concept of a Decision Tree Induction	K2	2	Lecture + discussion	BB
4.		Construct a decision tree.	K3	1	Lecture + discussion and In-class Assignment	BB
5.		Describe the methods for expressing an attribute test conditions.	K2	1	Lecture + discussion	BB
6.		Identify the measures for selecting the best split.	K2	1	Lecture + discussion	BB
7.		Apply the decision tree induction algorithm.	K3	2	Lecture + discussion and In-class Assignment	BB
		Total		**09**		

UNIT - IV

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO4	**Classification : Alternative techniques** Explain Bayes' Theorem	K1	2	Lecture+ discussion	BB
2.		Apply Naïve Bayesian Classification Algorithm	K3	2	Lecture + discussion and In-class Assignment	BB
3.		Explain the concept of Bayesian Belief Networks	K2	2	Lecture + discussion	BB
		Total		**06**		

UNIT - V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	C05	Illustrate the concept of Association Analysis	K2	2	Lecture + discussion	BB
2.		Explain Frequent Itemset generation process.	K2	2	Lecture + discussion	BB
3.		Discover Association Rules from the given data.	K3	2	Lecture + discussion and In-class Assignment	BB
4.		Present a Compact representation of frequent itemsets.	K1	1	Lecture	BB
5.		Use FP-Growth Algorithm for Association Analysis.	K3	2	Lecture + discussion and In-class Assignment	BB
		Total		**9**		

UNIT – VI

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO6	Explain the Concept of Cluster Analysis.	K2	1	Lecture + discussion	BB
2.		Describe different types of Clustering and Different Types of Clusters.	K2	2	Lecture + discussion	BB
3.		Apply k-means and its variants for clustering.	K3	3	Lecture + discussion and In-class Assignment	BB
4.		Describe k-means additional issues.	K1	1	Lecture	BB
5.		Describe k-means and different types of clusters.	K2	1	Lecture + discussion	BB
6.		Identify k-means strengths and weaknesses	K1	1	Lecture	BB
7.		Express k-means as an Optimization Problem.	K2	1	Lecture + discussion	BB
8.		Apply Basic Agglomerative Hierarchical Clustering.	K3	1	Lecture + discussion and In-class Assignment	BB
9.		Discuss other Specific Techniques for Agglomerative Hierarchical Clustering	K2	1	Lecture + discussion	BB
10.		Explain Traditional Density: Center-Based Approach.	K2	1	Lecture + discussion	BB
11.		Apply the DBSCAN Algorithm.	K3	1	Lecture + discussion and In-class Assignment	BB
12.		List strengths and Weaknesses of DBSCAN Algorithm	K1	1	Lecture	BB
		Total		**15**		

Total No. of classes: 60

Design and Analysis of Algorithms

LESSON PLAN

Academic Year:	2019-20	Programme:	B. Tech (CSE)
Year/Semester:	III/ II	Section:	A, B, C & D
Course Code:	R1632053/312	Course Name:	Design and Analysis of Algorithms

COURSE OUTCOMES:

After the completion of this course, students will be able to:

312.1 - Describe asymptotic notations and basic concepts of algorithms (**K2**)

312.2 - Apply divide and conquer paradigm to solve various problems (**K3**)

312.3 - Use greedy technique to solve various problems (**K3**)

312.4 - Apply dynamic programming technique to various problems (**K3**)

312.5 - Employ backtracking technique to various problems (**K3**)

312.6 - Apply branch and bound technique to various problems (**K3**)

TEXT BOOKS:

- Fundamentals of Computer Algorithms - Ellis Horowitz, SatrajSahni and Rajasekaran; 2/e; Universities Press.
- Design and Analysis of Algorithms - S Sridhar; Oxford.

REFERENCE BOOKS:

- Design and Analysis of Algorithms - Aho, Ullman and Hopcroft; Pearson Education.
- Introduction to Algorithms - T.H.Cormen, C.E.Leiserson, R.L.Rivest and C. Stein; 2/e; PHI

TARGETED PROFICIENCY AND ATTAINMENT LEVELS:

Course Outcome	Targeted Proficiency (Marks In %)	Targeted Attainment (Students In %)
312.1	60	70
312.2	60	70
312.3	60	65
312.4	60	60
312.5	60	70
312.6	60	60

LESSON PLAN

UNIT - 1 : Introduction						
S. No.	**Course Outcome**	**Intended Learning Outcome (ILO)**	**Knowledge Level of ILO**	**No. of Hours Required**	**Pedagogy**	**Teaching Aid**
		Dissemination of Dept. Vision & Mission, POs' and PSOs'				
1	312.1	Define Algorithm	K1	01	Lecture	BB
2		Express Pseudo Code for writing algorithms	K2	01	Lecture	BB
3		Explain Performance Analysis-Space Complexity, Time Complexity	K2	02	Lecture	BB
4		Illustrate Asymptotic Analysis	K2	01	Lecture	BB
5		Describe Asymptotic Notations- Big oh notation, Omega notation, Theta notation and Little oh notation	K2	01	Lecture	BB
6		Explain Amortized Analysis, Probabilistic Analysis	K2	01	Lecture	BB
			Total:	07		

UNIT - 2 : Divide and Conquer						
S. No.	**Course Outcome**	**Intended Learning Outcome (ILO)**	**Knowledge Level of ILO**	**No. of Hours Required**	**Pedagogy**	**Teaching Aid**
1	312.2	Describe Divide and Conquer strategy	K1	01	Lecture	BB
		Explain General method	K2	01	Lecture	BB
2		**Applications -** Explain Binary Search	K3	02	Lecture + Discussion	BB
3		Apply Divide and Conquer to Quick Sort	K3	02	Lecture + Discussion	BB
4		Apply Divide and Conquer to Merge Sort	K3	02	Lecture + Discussion	BB
			Total:	08		

UNIT - 3 : Greedy Method						
S. No.	**Course Outcome**	**Intended Learning Outcome (ILO)**	**Knowledge Level of ILO**	**No. of Hours Required**	**Pedagogy**	**Teaching Aid**
1	312.3	Describe Greedy method	K1	01	Lecture	BB
		Explain General method	K2	01	Lecture	BB

2		**Applications -** Solve Job Sequencing with deadlines	K3	02	Lecture + Discussion	BB
3		Solve Knapsack Problem	K3	02	Lecture + Discussion	BB
4		Explain Spanning Trees, Find Minimum Cost Spanning Trees	K3	01	Lecture + Discussion	BB
5		Solve Single Source Shortest Path Problem	K3	02	Lecture + Discussion	BB
			Total:	9		

UNIT - 4 : Dynamic Programming						
S. No.	**Course Outcome**	**Intended Learning Outcome (ILO)**	**Knowledge Level of ILO**	**No. of Hours Required**	**Pedagogy**	**Teaching Aid**
1	312.4	Describe Dynamic Programming	K1	01	Lecture	BB
2		Explain General method	K2	01	Lecture	BB
3		**Applications** - Apply Dynamic Programming to Matrix Chain Multiplication	K3	02	Lecture + Discussion	BB
4		Explain Optimal Binary Search Trees	K3	02	Lecture + Discussion	BB
5		Solve 0/1 Knapsack Problem	K3	02	Lecture + Discussion	BB
6		Solve All Pairs Shortest Path Problem	K3	02	Lecture + Discussion	BB
7		Demonstrate Travelling Sales Person Problem	K3	01	Lecture + Discussion	BB
8		Explain Reliability Design	K3	02	Lecture + Discussion	BB
			Total:	14		

UNIT - 5 : Backtracking						
S. No.	**Course Outcome**	**Intended Learning Outcome (ILO)**	**Knowledge Level of ILO**	**No. of Hours Required**	**Pedagogy**	**Teaching Aid**
1	312.5	Describe Backtracking strategy	K1	01	Lecture	BB
2		Explain General method	K2	01	Lecture	BB
3		**Applications:**Solve N-Queen Problem	K3	01	Lecture + Discussion	BB
4		Employ Backtracking to solve Sum of Subsets	K3	02	Lecture + Discussion	BB

		Problem				
5		Demonstrate Graph Coloring	K3	02	Lecture + Discussion	BB
6		Demonstrate Hamiltonian Cycles	K3	02	Lecture + Discussion	BB
			Total:	9		

UNIT - 6 : Branch and Bound						
S. No.	**Course Outcome**	**Intended Learning Outcome (ILO)**	**Knowledge Level of ILO**	**No. of Hours Required**	**Pedagogy**	**Teaching Aid**
1	312.6	Describe Branch and Bound technique	K1	01	Lecture	BB
2		Explain General method	K2	01	Lecture	BB
3		**Applications** - Solve Travelling Sales Person problem	K3	02	Lecture + Discussion	BB
4		Apply Branch and Bound to 0/1 Knapsack problem	K3	03	Lecture + Discussion	BB
5		Explain LC Branch and Bound solution	K3	03	Lecture + Discussion	BB
6		Explain FIFO Branch and Bound solution	K3	03	Lecture + Discussion	BB
			Total:	13		

Total Number of Hours: 60

Software Testing Methodologies

Academic Year: 2019-20 Programme: B.Tech
Year/ Semester: III/II Section: A,B , C & D
Name of the Course: STM Course Code:R1632054/C-313

S.NO	**COURSE OUTCOMES** (After completion of the course, The Learner is able to)	KNOWLEDGE LEVEL
CO1	Describe the basic testing procedures.	K2
CO2	Apply data flow testing techniques.	K3
CO3	Discuss Domain testing and Regular expressions.	K2
CO4	Demonstrate Syntax testing and Logic based Testing.	K3
CO5	Explain Transition testing and Graph matrices.	K2
CO6	Apply software testing methods and modern software testing tools for their testing projects.	K3

Course Outcomes:
:

CO	CO1	CO2	CO3	CO4	CO5	CO6
Targeted Level of Proficiency (Marks in %)	65%	60%	65%	60%	65%	60%
Targeted Level Of Attainment (Student in %)	60%	50%	60%	50%	60%	50%

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO 1	Introduction to OBE, Dissemination of Vision, Mission of the Dept.and PEOs,POs & PSOs of the Programme.		1	Lecture	BB
2.		Describe the Purpose of Testing	K1	1	Lecture	BB
3.		Describe Dichotomies	K1	2	Lecture	BB
4.		Explain Model for Testing	K2	1	Lecture	BB
5.		Illustrate Consequences of Bugs	K2	1	Lecture	BB
6.		Explain Taxonomy of Bugs	K2	1	Lecture	BB
7.		Describe the basic concepts of Path testing	K1	1	Lecture	BB
8.		Explain Path Predicates And Achievable Paths,	K2	2	Lecture	BB
9.		Explain Path Sensitizing, Path Instrumentation, Application of Path Testing.	K2	2	Lecture	BB
Total :12hrs						

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO 2	Describe Transaction Flows	K2	1	Lecture	BB
2.		Illustrate Transaction Flow Testing Techniques.	K3	2	Lecture	BB
3.		Explain Basics of Dataflow Testing, .	K2	2	Lecture	BB
4.		Describe Strategies in Dataflow Testing,	K2	2	Lecture	BB
5.		Develop Application Of Dataflow Testing	K3	2	Lecture	BB
Total :09hrs						

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO3	Describe Domains and Paths	K1	1	Lecture	BB
2.		Explain Nice & Ugly Domains	K2	1	Lecture	BB
3.		Explain Domain testing	K2	1	Lecture	BB
4.		Describe Domains and Interfaces Testing	K2	1	Lecture	BB
5.		Discuss Domains and Testability	K2	1	Lecture	BB
6.		Illustrate Path Products & Path Expression	K2	1	Lecture	BB
7.		Describe Reduction Procedure, Applications	K2	2	Lecture	BB
8.		Explain Regular Expressions & Flow Anomaly Detection	K2	2	Lecture	BB
Total :10hrs						

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO 4	Explain Syntax Testing Why, What and How	K2	1	Lecture	BB
2.		Develop A Grammar for formats and Test Case Generation	K3	2	Lecture	BB
3.		Construct and Implementation , Application and Testability Tips	K3	2	Lecture	BB+ICT
4.		Explain Overview and Decision Tables	K2	2	Lecture	BB
5.		Develop Path Expressions, KV Charts, and Specifications.	K3	3	Lecture	BB+ICT
Total :10hrs						

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
	CO5	Describe State Graphs, Good & Bad State Graphs	K1	2	Lecture	BB+ICT
		Explain State Testing, and Testability Tips.	K2	2	Lecture	BB+ICT
		Discuss Motivational overview, matrix of graph	K2	2	Lecture	BB
		Explain relations, power of a	K2	2	Lecture	BB

		matrix.				
		Discuss Node reduction algorithm	K2	2	Lecture	BB
Total :10hrs						

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
	CO6	Explain Testing, Automated Testing and Concepts of Test Automation	K2	1	Lecture	BB+ICT
		Illustrate list of tools like Win runner, Load Runner, Jmeter	K3	1	Lecture	BB+ICT
		Demonstrate About Win Runner ,Using Win runner and Mapping the GUI,	K3	2	Lecture	BB+ICT
		Develop Recording Test, Working with Test and Enhancing Test,	K3	2	Lecture	BB+ICT
		Develop Checkpoints, Test Script Language and Putting it all together, Running and Debugging Tests, Analyzing Results,	K3	2	Lecture	BB+ICT
		Examine Batch Tests, Rapid Test Script Wizard.	K3	1	Lecture	BB+ICT
Total :09hrs						

Total Hours: 60

Cyber Security

Academic Year: 2019-20 **Programme:** B.Tech

Year/ Semester: III/II **Section:** A,B,C,D

Name of the Course: Cyber Security **Course Code: R163205C/C-314B**

Course Outcomes (Along with Knowledge Level):

After Completing the course Student will be able to:

S.No	Course Outcomes	Knowledge
CO1	Demonstrate about Cyber crime	**K2**
CO2	Explain about Cyber criminals	**K2**
CO3	Express about Cyber crime in mobile devices	**K2**
CO4	Discuss about the Tools and methods used for overcome Cyber crime	**K3**
CO5	Describe about Cyber Laws	**K2**
C06	Explain about Computer Forensics.	**K2**

TEXT BOOKS:

1. CyberSecurity:UnderstandingCyberCrimes,ComputerForensicsandLegalPerspectives, NinaGodbole, SunitBelapure, Wiley.
2. Principles ofInformationSecurity, MichealE.Whitmanand Herbert J.Mattord, Cengage Learning.

REFERENCES:

1.InformationSecurity,MarkRhodes, Ousley, MGH.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
CO1	60	60
CO2	60	60
CO3	60	60
CO4	60	60
CO5	60	60
CO6	60	60

CO1:UNIT-I

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Discuss Blooms Taxonomy	K1 and K2	1	LECTURE	BB+ICT
2		Explain Cyber Crime	K2	1	LECTURE	BB
3		Define about Cyber Crime Describe originsof the Word	K1 and K2	1	LECTURE	BB
4		Define Information Security and also Discuss about Cybercrime andInformationSecurity	K1 and K2	1	LECTURE	BB
5		Express whoareCybercriminals?	K2	1	LECTURE	BB
6		ClassificationsofCybercrimes	K2	1	LECTURE with Discussion	BB
7		Explain about Cybercrime:TheLegalPerspectives	K2	1	LECTURE with Discussion	BB
8		Explain about Cybercrimes:AnIndian Perspective. Cybercrime andtheIndianITA 2000	K2	1	LECTURE	BB
9		Describe the Global Perspective on Cybercrimes.	K2	1	LECTURE	BB
10		Discuss about Cybercrime Era: Survival Mantra for the Netizens.	K2	1	LECTURE	BB
		TOTAL		**10**		

CO2: UNIT-II

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	List about the planning of criminals	K1	1	LECTURE	BB
2		Identify how criminalsPlantheAttacks	K2	2	Discussion	ICT+BB
3		Explain about SocialEngineering	K2	1	Discussion	ICT+BB
4		Discuss about Cyber stalking	K2	1	Discussion	ICT+BB
5		Explain about Cyber cafe andCybercrimes	K2	2	Discussion	ICT+BB
6		Discuss about Botnets:TheFuelforCybercrime	K2	1	Discussion with Lecture	BB
7		AttackVectorCloud Computing	K2	2	Discussion with Lecture	ICT+BB
		TOTAL		**10**		

CO3: UNIT-III

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Define about wireless devices	K1	1	LECTURE	BB
2		Explain the Proliferationof Mobile andWirelessDevices	K2	1	LECTURE	BB
3		DiscussthetrendsinMobility	K2	1	LECTURE	BB+ICT
4		Demonstrate Credit Card FraudsinMobileandWirelessComputingEra	K2	1	Lecture	ICT
5		Explain the Security ChallengesPosed by MobileDevices	K2	1	LECTURE	BB
6		Describe the Registry SettingsforMobileDevices	K2	1	Discussion	BB+ICT
7		Define Authentication and Explain the AuthenticationServiceSecurity	K1 and K2	1	ICT	BB
8		Discuss AttacksonMobile/Cell Phones	K2	1	Lecture	ICT
9		Explain about MobileDevicesSecurityImplicationsforOrganizations	K2	1	Lecture	BB
10		Discuss about OrganizationalMeasuresforHandlingMobile	K2	1	Lecture	ICT
11		Express the OrganizationalSecurityPoliciesandMeasuresinMobileComputingEra, Laptops.	K2	1	Lecture with discussion	BB
		TOTAL		**1**		

CO4: UNIT-IV

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Describe about Proxy ServersandAnonymizers	K2	1	LECTURE	BB
2		Explain about Phishing, PasswordCracking	K2	1	Lecture with Discussion	BB
3		Name theKeyloggersandSpywares	K1	1	Lecture with Discussion	BB+ICT
4		Describe about VirusandWorms, TrojanHorsesandBackdoors	K2	1	LECTURE	BB
5		Discuss about Steganography,DoSandDDoS Attacks	K2	2	Discussion	BB

6		Explain about SQLInjection,BufferOverflow	K2	1	Discussion	BB
7		Identify the AttacksonWirelessNetworks	K2	1	LECTURE	BB
8		Phishing andIdentity Theft	K2	1	LECTURE	BB
		TOTAL		**9**		

CO5: UNIT-V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	Describe about the IndianContext of cyber laws, theIndianITAct	K2	1	LECTURE	BB
2		Summarize the ChallengestoIndianLawandCybercrimeScenarioinIndia	K2	1	LECTURE	BB
3		Discuss about the ConsequencesofNotAddressingtheWeaknessinInformation Technology Act	K2	1	LECTURE	BB
4		Discuss about the Digital Signatures and the Indian IT Act	K2	1	LECTURE	BB
5		Explain about Information Security Planning and Governance.	K2	1	Discussion	BB+ICT
6		Discuss the Information Security Policy Standards, Practices.	K2	1	LECTURE	BB
7		Describe theinformationSecurityBlueprint,Securityeducation.	K2	1	LECTURE	BB
8		Explain about training andawareness program, ContinuingStrategies.	K2	1	LECTURE	BB+ICT
		TOTAL		**8**		

CO6: UNIT-VI

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO6	Define Computer forensics	K1	1	LECTURE	BB+ICT
2		Discuss the HistoricalBackgroundofCyber forensics, DigitalForensics Science	K2	1	LECTURE	BB
3		Describe about The NeedforComputer Forensics, Cyber forensics and Digital Evidence	K2	1	LECTURE	BB

4		Discuss about forensics Analysis of E-Mail.	K2	1	LECTURE and Discussion	BB+ICT
5		Explain about DigitalForensicsLifeCycle,ChainofCustodyConcept	K2	1	LECTURE	BB+ICT
6		Describe about networkForensics, ApproachingaComputerForensicsInvestigation	K2	1	LECTURE	BB
7		Discuss about computerForensicsandSteganography	K2	1	LECTURE with Discussion	BB
8		Explain about relevanceofthe OSI 7LayerModeltoComputerForensics	K2	1	LECTURE	BB+ICT
9		Describe about the ForensicsandSocialNetworkingSites:TheSecurity/Privacy Threats	K2	1	LECTURE	BB
10		Explain about computerForensicsfromCompliancePerspective	K2	1	LECTURE with Discussion	BB
11		Describe about the challengesinComputer Forensics, Special Tools and Techniques	K2	1	LECTURE	BB+ICT
12		Discuss about forensics Auditing, Ant forensics	K2	1	LECTURE	BB
		Total		**12**		

Total Number of Hours= 60

Network Programming Lab

Academic Year: 2019-20 **Programme**:B.Tech.

Year/Sem: III/II

Name of the Course: Network Programming Lab **Course Code**: **C-316**

COURSE OUTCOMES:

After completion of this course the students will be able to

CO.1	Construct C Programs Which Use Socket API. (K2)
CO.2	Construct Client/Server Architecture in Application Development. (K2)
CO.3	Use TCP And UDP Based Sockets and Their Differences. (K3)
CO.4	Develop Unix System Internals Like Socket Files, IPC Structures. (K3)
CO.5	Develop Reliable Servers Using Both TCP And UDP Sockets. (K3)

Software Testing LAB

LESSON PLAN

Academic Year: 2019-20	**Programme**: B.Tech
Year/Sem: III/II	
Name of the Course: Software Testing LAB	**Course Code**: **R1632057/C-317**

COURSE OUTCOMES:

After completion of this course, the students will be able to

C-317.1	Describe the testing process.(K2)
C-317.2	Apply different software testing techniques.(K3)
C-317.3	Construct test cases by understanding test suite management and software quality management.(K3)
C-317.4	Demonstrate modern software testing tools and testing of Object Oriented Software and Web based software.(K3)

Data Warehousing and Mining Lab

Batch: 2017-21

Academic Year : 2019-20 **Programme: B.Tech**

Year/ Semester: III /II **Section: A,B,C,D**

Name of the Course: Data Warehousing and Mining Lab

Course Code: R1632058 / 318

Course Outcomes:

After completion of this course, the student will be able to

C318.1	Demonstrate Data Pre-processing techniques.[K3]
C318.2	Demonstrate Association Rule Mining techniques.[K3]
C318.3	Demonstrate Classification techniques. [K3]
C318.4	Demonstrate the Clustering techniques. [K3]

Intellectual property

Academic Year: 2019-20 **Programme: B.Tech**

Year/ Semester: III/II **Section: A,B,C & D**

Name of the Course: Intellectual property Right and patens

Course Code:C315/ R1632049

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO1. Understand the principles, function and basic legal rules of IP Law.[K1]

CO2. Explain copyright author(s) rights, licensing and international copyright laws.[K2]

CO3.Understand patentability requirements, ownership issues, and protection of patent rights.[K2]

CO4. Understand law of trademarks, registration process, infringement of trademarks. [K2]

CO5. Understand tradesecret laws and protection of tradesecrets. [K1]

CO6. Understand types of cybercrimes and information technology act. [K1]

TEXT BOOKS:

T1. “Intellectual Property” ,Deborah E.Bouchoux, Cengage learning

T2. "Intellectual Property", Richard Stim, Cengage Learning, New Delhi.

T3. Cyber Law Texts & Cases, South-Western’s Special Topics Collections

REFERENCE BOOKS:

R1. Kompal Bansal & Parishit Bansal "Fundamentals of IPR for Engineers",BS Publications

R2. Prabhuddha Ganguli: “ Intellectual Property Rights” Tata Mc-Graw –Hill, New Delhi

R3. Intellectual Property Rights, Dr. A. Srinivas. Oxford University Press, New Delhi.

R4. M.Ashok Kumar and Mohd.Iqbal Ali: “Intellectual Property Right” Serials Pub

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
CO1	60	70
CO2	60	70

CO3	60	70
CO4	60	70
CO5	60	70
CO6	60	70

CO 1

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Dissemination of Vision, Mission of the Dept.and PEOs,Pos,& PSOs of the Programme			Lecture	BB
2		Define Intellectual Property law and types of Intellectual Property	K1	1	Lecture	BB
3		List the agencies Responsible for Intellectual property Registration	K1	1	Lecture	BB
4		Identify Infringement regulatory, Over use or Misuse of Intellectual Property Rights	K1	1	Lecture	BB
5		List the emerging areas of IPR - Layout Designs and Integrated Circuits.	K1	1	Lecture	BB

CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	State the Principles, Subject Matters of Copyright	K1	1	Lecture	BB

2		Explain Rights Afforded by Copyright Law, Right to Prepare Derivative Works, Rights of Distribution , Rights of performers, Copyright Ownership, Transfer and Duration	K2	1	Lecture with Discussion	BB
3		Explain Copyright Formalities and Registration	K2	1	Lecture with Discussion	BB
4		Discuss infringement of Copyright, International Copyright Law, Semiconductor Chip Protection Act	K2	1	Lecture with Discussion	BB

CO 3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	State Patent requirements, Rights under Patent law and Limitations, Ownership and Transfer	K1	1	Lecture	BB
2		Explain Patent Application Process and Granting of Patent	K2	1	Lecture with Discussion	BB
3		Discuss Patent Infringement and Litigation, Double Patenting, Patent Searching	K2	1	Lecture with Discussion	BB
4		Enumerate International Patent Law, Patent Cooperation Treaty New developments in Patent Law	K1	1	Lecture	BB

CO 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Explain Trade Mark registration process, Post registration procedures	K2	1	Lecture with Discussion	BB
2		Discuss Trademark maintenance, Transfer of rights, Inter parties Proceedings	K2	1	Lecture with Discussion	BB
3		Discuss Infringement, Dilution of ownership of Trade Mark, Likelihood of confusion	K2	1	Lecture with Discussion	BB
4		Identify Trade Mark claims, Trade Marks Litigation, International Trade Mark Law	K1	1	Lecture	BB

CO 5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Define Tradesecret, Maintaining Trade Secret by Physical Security, Employee Access Limitation, Employee Confidentiality Agreement	K1	1	Lecture	BB
2		State Trade Secret Law, Unfair Competition, Breach of Contract	K1	1	Lecture	BB
3		Outline Trade Secret Litigation,	K1	1	Lecture	BB

CO 6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Outline Cyber Law , Information Technology Act	K1	1	Lecture	BB
2		Describe Cyber Crime and E-commerce,Data Security , Confidentiality and Privacy	K1	1	Lecture	BB
3		List International aspects of Computer and Online Crime	K1	1	Lecture	BB

Total No. of Classes: 22

English Communication Skills

Academic Year: 2018-19 **Programme: B.Tech**

Year/ Semester: III/ II **Section: A,B,C & D**

Name of the Course: ECS

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course students will be able to

CO 1. Comprehend the errors in the sentence formation. [K2]

CO 2. Show case required qualifications for his/her career.[K3]

CO 3. perceivethe possible ways of participating in the discussion .[K3]

CO 4.Appraise oneself professionally in the interview[K3]

CO 5 Generate ideas and showcase their skills .[K6]

CO 6. organizetheir thoughts in the process of reading and writing different concepts. [K3]

REFERENCE BOOKS:

R1. GRE - Barons

R2. CAT - Muneer

R3. Essential Grammar in Use – RAYMOND MURPHY

R4. English for Professional Students – S.S.Prabhakar

R.5. General English for Competitive Examination –

R.6. Word Power Made Easy Handy – Dr.ShaliniVerma

R.7. A Practical English Grammar – A.J.Thomson

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	50	60
2	50	60
3	50	60
4	50	60
5	50	60
6	50	60

CO 1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Construct appropriate sentences by identifying the errors in Nouns, Pronouns, verbs.	K2	1	Lecture	BB
2		Construct appropriate sentences by identifying the errors in adverbs, adjectives, conjunctions and prepositions	K2	1	Lecture with Discussion	BB

CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Understands the significance of building one's profike	K3	2	Lecture	Hand outs/clippings
2		Can prepare career profile	K3	2	Team work	A.V
3						

CO 3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Acquire various techniques of presenting one's views in a polite way in Group Discussion	K3	2	Lecture/A.v/Team work	BB
		Hands on experience of real time discussions	K3	2	Discussion	--

Co4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Understands the intricacies of better performance in the interviews	K3	2	Lecture/Discussion	BB
2		Experiential learning through mock drives	K3	2	Individual performance	--

CO5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Fabrication of a story based on specific words	K6	2	Realia/Discussion	BB
2		Create Captions/Tag lines	K6	2	Team work	A.V.

CO6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Apply logic and answer Reading comprehension questions	K6	2	Realia/Discussion/ Team work	BB/AV

Total No. of Classes: 14

Aptitude

Academic Year: 2019-20 **Programme: B.Tech**
Year/ Semester: III/II **Section: A,B,C & D**
Name of the Course: Aptitude **Course Code:APT**

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:
CO 1. Identify the easiest and best possible way of solving problems in the area of Cubes and Data Analysis. [K 3]

CO 2. Investigate the different types of logics involved in Analytical Reasoning, Logical Reasoning & arithmetical Reasoning. [K 4]

CO 3. Observe the common traps in the questions and errors likely to be made from the concepts of Data Sufficiency, Permutations-Combinations and Probability. [K 2]

CO 4. Improve problem solving skills through the concepts of Time & Work, Time & Distance and Simple Interest & Compound Interest. [K 3]

CO 5. Analyze appropriate methods of logical thinking on Number System, Areas & Volumes. [K 4]

TEXT BOOKS:
T1. AWork book on Aptitude. Part – II.

REFERENCE BOOKS:
R1. Quantitative aptitude by R.S.Agarwal.
R2. Shortcuts on Aptitude by M.Tyra.
R3. Various Study Center Materials.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	50	50
2	50	50

3	50	50
4	50	50
5	50	50

CO 1

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Data Analysis helps and to determine if the given information is sufficient to take decision	K3	4	Lecture	BB
2		Cubes is one of the top competitive subject which will really want to learn and to be expert.			Lecture	BB

CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Analytical Reasoning, this approach is usually associated with finding evidence to either support or reject hypotheses	K4	6	Lecture	BB

CO 3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Describe Engineering defective parts from probability	K2	3	Lecture	BB
6		Differentiate and Explain permutations & combinations.	K2	3	Lecture with Discussion	BB

CO 4

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Improve problem solving skills through the concepts of Time & Work, Time & Distance and Simple Interest & Compound Interest.	K3	8	Lecture	BB

CO 5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Analyze appropriate methods of logical thinking on Number System, Areas & Volumes.	K4	4	Lecture	BB

Total No. of Classes: 28